Retrieval-Guided Input-Space Adaptation for Frozen Time-Series Predictors

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Abstract

Clinical predictors degrade across hospitals, but regulatory and hardware constraints often forbid changing the validated model. We introduce RETRADAPTER, a retrieval-guided input-space adapter that adapts the data, not the model, while the downstream predictor stays strictly frozen. Most domain adaptation methods assume source labels are unavailable and the encoder is trainable. These assumptions fit many settings, but a complementary and under-studied regime arises when the target predictor's weights cannot be changed and source labels are available, a common situation in SaMD transfer, firmware-embedded models, and edge deployment. In this regime, alignment losses cannot propagate through a frozen encoder and classical DA is structurally blocked. RETRADAPTER encodes source windows, retrieves k nearest target exemplars per timestep, and fuses them via cross-attention before decoding. Across five ICU tasks (eICU, HiRID $\rightarrow$ MIMIC-IV) and five AdaTime benchmarks, a single method strictly outperforms every comparable baseline: all four frozen-backbone DA methods, an affine statistics baseline, end-to-end fine-tuning of the predictor, three end-to-end DA methods on the clinical classification tasks, and all eleven published end-to-end methods on AdaTime under that benchmark's own training protocol. Adapted predictions exceed the target-native LSTM on two of five tasks and the source-native LSTM on four of five, with AKI reaching 91.1 AUROC ($+16.1$ AUCPR over target-native).

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1. Introduction

Clinical prediction models trained at one hospital degrade when deployed at another due to differences in equipment calibration, clinical protocols, and patient populations. On the YAIB benchmark (van de Water et al., 2024), a MIMIC-trained LSTM sepsis predictor drops from 82.0 to 71.6 AUROC when evaluated on eICU data, and an acute kidney injury (AKI) model falls from 89.7 to 85.6, losses that materially affect clinical utility. Across five tasks spanning both classification and regression, cross-domain degradation ranges from 4 to 10 AUROC points, undermining the reliability that bedside deployment demands. Retraining on the new domain is the standard remedy, but natively trained target-domain models perform only marginally better than the cross-domain frozen predictor, and modifying a validated predictor triggers regulatory re-certification under the FDA's Software as a Medical Device framework (U.S. Food and Drug Administration, 2021). The same constraint arises outside healthcare: firmware-embedded models on wearable devices, industrial IoT sensors, and edge-deployed sleep-staging systems cannot be retrained per-user or per-deployment site. The result is a practical impasse: the model cannot be changed, yet it degrades on new data.

Domain adaptation (DA) methods (Ganin et al., 2016; Sun & Saenko, 2016; Wilson et al., 2020) align feature representations through a jointly trained encoder, but a frozen encoder blocks gradient-based alignment. Frozen-model alternatives from vision and NLP (Jia et al., 2022; Qiu et al., 2026) prepend learnable tokens or apply handcrafted transforms, which lack the capacity for clinical time series where dozens of features shift simultaneously. We propose *input-space domain adaptation*: adapt the data, not the model. Classical DA has made strong progress under the assumption that source labels are unavailable and the encoder is trainable. Our focus is complementary: a deployment setting in which the target predictor is locked after certification and source labels are available. Under this regime, task loss on the source can supervise an adapter directly. Fully unlabeled source–target adaptation remains outside our scope, and we make no claim to improve on it; we target the frozen-predictor regime, where existing methods are structurally blocked. RETRADAPTER encodes source and target windows into

a shared latent space, retrieves k nearest target exemplars from a memory bank, and fuses them via cross-attention before decoding, while the predictor remains frozen. Prior input-space methods apply handcrafted or unconditional transforms to single modalities such as images or univariate foundation-model inputs. RETRADAPTER learns instance-conditional, retrieval-guided transformations on multivariate clinical time series with variable-length stays while keeping the predictor strictly frozen, a combination we demonstrate across five clinical tasks, three downstream architectures, and five non-clinical benchmarks.

Contributions.

- **Frozen predictors can be adapted through input-space learning.** When alignment gradients cannot reach a locked encoder, a retrieval-guided adapter trained on source task labels can reshape the input instead. The predictor's weights and regulatory certification stay intact, yet adapted predictions beat four frozen-backbone DA methods (DANN, Deep CORAL, CoDATS, ACON (Liu et al., 2024)) by 1.3–3.7×, an affine statistics-only baseline by a wide margin, and end-to-end fine-tuning of the predictor by +0.007 to +0.032 AUROC, across five clinical tasks and two source domains (eICU, HiRID → MIMIC-IV). To our knowledge, no prior work addresses per-timestep frozen-predictor DA on clinical EHR, a setting that combines a strictly locked downstream model with instance-level target retrieval.
- **Input-space adaptation can exceed natively trained models.** On AKI and LoS, the adapted eICU signal surpasses both the eICU-native and MIMIC-native LSTMs (AKI 91.1 vs 89.7 AUROC, +16.1 AUCPR reaching 72.9 from 56.8; LoS 37.2 h vs 40.6 h MAE). Mortality (85.7 vs 85.5) and Sepsis (77.8 vs 74.0) also surpass the eICU-native LSTM.
- **The approach is general across domains, architectures, and modalities.** On five non-clinical AdaTime benchmarks (wearable, EEG, industrial) with a frozen 1D-CNN, the adapter beats all eleven published end-to-end DA methods that retrain the full model (+2.6 mean Macro-F1), trained under AdaTime's exact recipe (Appendix B). Separately, an LSTM-trained adapter transfers zero-shot to frozen GRU and TCN predictors (47–86% of LSTM gain retained). Input-space adaptation is not specific to clinical time series: the same architecture applies whenever the downstream predictor must remain frozen.

2. Methods

2.1. Problem Setting

Let $\mathcal{X}_S \sim P_S$ denote source-domain clinical time series (eICU or HiRID) and $\mathcal{X}_T \sim P_T$ denote the target domain (MIMIC-IV). A predictor $f_T : \mathbb{R}^{T\times F} \to \mathbb{R}^{T\times C}$, trained on

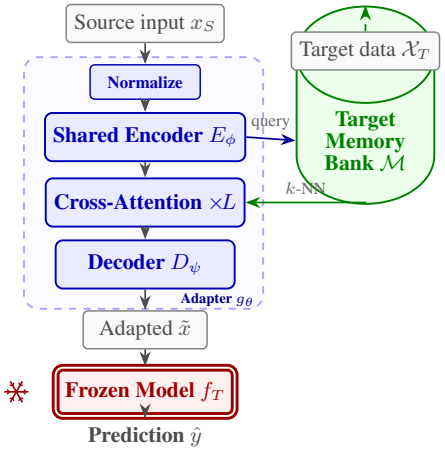

Figure 1. The RETRADAPTER architecture. Source input is encoded, matched to target exemplars by k-NN, fused through cross-attention, and decoded for the frozen downstream model.

$\mathcal{X}_T$, is *entirely frozen*: all parameters satisfy $\nabla_\phi f_T = 0$. We learn an adapter $g_\theta : \mathbb{R}^{T\times F} \to \mathbb{R}^{T\times F}$ that maps source inputs into the target input space so that the composite system $f_T \circ g_\theta$ performs well on source labels y_S:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{task}}\big(f_T(g_\theta(x_S)),\, y_S\big) + \lambda_{\text{fid}}\, \mathcal{L}_{\text{fid}}\big(g_\theta(x_S),\, x_S\big) + \lambda_{\text{range}}\, \mathcal{L}_{\text{range}}\big(g_\theta(x_S)\big), \quad (1)$$

where $\mathcal{L}_{\text{task}}$ is cross-entropy (classification) or MSE (regression) evaluated through the frozen predictor, $\mathcal{L}_{\text{fid}}$ is an MSE fidelity penalty regularizing toward identity, and $\mathcal{L}_{\text{range}}$ penalizes outputs outside the physiological range observed in $\mathcal{X}_T$. Equation (1) governs Phase 2 (adaptation training); Phase 1 uses a separate reconstruction objective on target data only (§2.3). The frozen constraint reflects deployment reality: under the FDA SaMD framework, validated predictors cannot be modified post-certification (U.S. Food and Drug Administration, 2021).

2.2. Retrieval-Guided Architecture

The RETRADAPTER architecture (Figure 1) comprises four components: a shared encoder, a target-domain memory bank, cross-attention fusion blocks, and a decoder.

Shared encoder and memory bank. A shared encoder E_ϕ maps both source and target windows to a latent space of dimension d_{latent}: $z_s = E_\phi(x_s) \in \mathbb{R}^{T\times d_{\text{latent}}}$. Each encoder and decoder layer is an *axial attention block* that applies self-attention along both temporal and feature dimensions, so the model handles per-feature dynamics and cross-feature interactions without concatenating the full $T \times F$ sequence. All target-domain training data is encoded and segmented into temporal windows of size W timesteps, each mean-pooled into a compact descriptor (Appendix B). The resulting memory bank $\mathcal{M} = \{E_\phi(x_T^j)\}_{j=1}^{|\mathcal{X}_T|}$ resides on GPU for

fast retrieval and is rebuilt every R epochs as the encoder's representations evolve.

Per-timestep k-NN retrieval. For each source timestep t, a query is formed by mean-pooling source latents over a backward-looking window $[\max(0, t-W+1),\, t]$, preserving causal consistency. Learned importance weights $\sigma(\mathbf{w}) \in [0,1]^{d_{\text{latent}}}$ modulate a weighted Euclidean distance over latent dimensions, and the k nearest target windows are retrieved, with k selected by grid search (Appendix B). This per-timestep design matches local physiological states rather than full trajectories and outperforms wider retrieval windows in all tasks.

Cross-attention fusion. Retrieved target context is integrated through L stacked cross-attention blocks. Let $\mathcal{N}_k(z_s^t)$ denote the k nearest target latents for timestep t; each block computes:

$$z_{\text{cross}}^t = \text{CrossAttn}\big(Q{=}z_s^t,\ K{=}V{=}\mathcal{N}_k(z_s^t)\big) + z_s^t, \quad (2)$$

followed by causal self-attention and a feed-forward network, both with residual connections and pre-LayerNorm. Cross-attention allows selective, feature-level borrowing from target exemplars, so the model attends to different features from different neighbors rather than uniformly interpolating as in kNN-MT (Khandelwal et al., 2021). The decoder D_ψ maps the fused representations back to input space:

$$\tilde{x} = D_\psi(z_{\text{cross}}) \in \mathbb{R}^{T\times F}. \quad (3)$$

The cross-attention depth L is selected per task (Appendix B).

2.3. Training

Training follows a two-phase protocol. **Phase 1** (autoencoder pretrain): E_ϕ and D_ψ are trained on target data $\mathcal{X}_T$ only with a reconstruction objective; ablations (Appendix E) show this phase is critical. **Phase 2** (adaptation): end-to-end training on source data through the frozen predictor with the objective of Equation (1), where $\mathcal{L}_{\text{task}}$ supplies the primary gradient through f_T and $\mathcal{L}_{\text{fid}}$, $\mathcal{L}_{\text{range}}$ regularize the output; the memory bank is rebuilt every R epochs as representations evolve.

Before adaptation, source features undergo cross-domain normalization: an affine transform matching per-feature mean and variance to target statistics, computed once from training data. Attention is causal for per-timestep tasks (AKI, Sepsis, LoS) and bidirectional for per-stay tasks (Mortality, KF), preventing information leakage across time. AdaTime datasets (HAR, HHAR, WISDM, SSC, MFD) present fixed-length windows with a single label per window and use bidirectional attention throughout.

3. Experiments

3.1. Setup

Clinical benchmarks. We evaluate on three ICU databases preprocessed via the YAIB benchmark (van de Water et al., 2024): eICU (source, 200,859 stays across 208 US hospitals), MIMIC-IV (target, 73,181 stays), and HiRID (second source, 33,905 stays). Five tasks span classification and regression: *Mortality24* (per-stay binary), *AKI* (per-timestep, KDIGO Stage $\geq$1, 12% positive), *Sepsis* (per-timestep, Sepsis-3, 1.1% prevalence), *Length of Stay* (LoS; per-timestep regression), and *Kidney Function* (KF; per-stay regression, serum creatinine). Dataset and feature details are in Appendix A. The frozen predictor is a MIMIC-IV-trained LSTM from YAIB, used identically for all methods.

General time-series benchmarks. We also evaluate on five datasets from the AdaTime benchmark (Ragab et al., 2023): three wearable (HAR, HHAR, WISDM), one EEG (SSC), and one industrial (MFD) dataset. The frozen backbone is AdaTime's standard pretrained 1D-CNN; every AdaTime split ships with labeled source data.

Baselines. Under the frozen-predictor constraint we compare against four DA methods, DANN (Ganin et al., 2016), Deep CORAL (Sun & Saenko, 2016), CoDATS (Wilson et al., 2020), and ACON (Liu et al., 2024), all adapted so the predictor's weights never move. We add two reference baselines: an affine statistics-only renormalization (negative control, no learning), and end-to-end fine-tuning of the LSTM (upper bound; violates our frozen constraint). As a further stress test we also run three end-to-end DA methods that retrain the full LSTM on each clinical classification task, CLUDA (Ozyurt et al., 2023), RAINCOAT (He et al., 2023), and ACON (Liu et al., 2024); results are in Appendix B. We also compare against all six natively trained architectures from YAIB (LSTM, GRU, TCN, Transformer, LogReg, LGBM), the in-domain ceiling. For AdaTime, we compare against all eleven published end-to-end DA methods from Ragab et al. (2023) (full list in Appendix B), plus CLUDA (Ozyurt et al., 2023) and RAINCOAT (He et al., 2023), all run under AdaTime's exact training recipe (Appendix B). We report AUROC ($\times$100) for classification and MAE for regression, with bootstrap 95% CIs (500 replicates); all $p < 0.001$. Across every comparable setting (frozen-backbone DA, affine renormalization, end-to-end fine-tuning of the LSTM, end-to-end DA on the clinical classification tasks (Appendix B), and end-to-end DA on AdaTime), RETRADAPTER strictly dominates.

3.2. Clinical Task Results

RETRADAPTER surpasses all four frozen-backbone DA baselines by 1.3–3.7$\times$, beats end-to-end fine-tuning on the three classification tasks ($+0.007$ to $+0.032$ AUROC), and

Table 1. Domain adaptation results across five clinical tasks. Classification: AUROC ($\times 100$, ↑); regression: MAE (↓). Baselines grouped by constraint. † surpasses eICU-native LSTM; ‡ surpasses MIMIC-native LSTM. ⋆ task-specific loss variant (Appendix E). Best method sharing our frozen-predictor constraint in **bold**. Bootstrap 95% CIs (500 replicates) shown for classification; regression CIs in Appendix C.2. Fine-tune LSTM modifies the predictor and is included as a non-frozen reference only.

	CLASSIFICATION (AUROC ↑)			REGRESSION (MAE ↓)	
Method	Mortality	AKI	Sepsis	LoS (h)	KF (mg/dL)
Frozen predictor (our constraint)					
Frozen baseline (source only)	80.79	85.58	71.59	42.5	0.403
Affine statistics-only	79.79	84.46	70.87	—	—
DANN	84.38	88.74	73.23	—	—
Deep CORAL	84.53	88.66	73.26	—	—
CoDATS	84.31	86.84	71.22	—	—
ACON	84.48	88.62	71.08	—	—
RETRADAPTER (eICU→MIMIC)	**85.74** †	**91.14** †‡	**77.76** †	**37.2** †‡⋆	**0.292** ⋆
95% CI	[84.6, 86.5]	[91.1, 91.2]	[77.4, 78.2]		
Fine-tune LSTM (end-to-end)	84.61	90.48	74.59	—	—
Reference: natively trained target-domain models					
eICU-native LSTM	*85.5*	*90.2*	*74.0*	*39.2*	*0.28*
MIMIC-native LSTM	*86.7*	*89.7*	*82.0*	*40.6*	*0.28*

Table 2. AdaTime benchmark (Macro-F1 $\times 100$). Our frozen-backbone adapter (only adapter parameters trained) vs. best published end-to-end method per dataset (full model retrained). Mean±std over 5 seeds; all $p < 0.0001$.

Dataset	Domain	Source	Best E2E	Ours	Δ
HAR	Wearable	80.0	93.7[a]	**94.1**±0.0	+0.4
HHAR	Wearable	56.5	84.5[b]	**87.0**±0.7	+2.5
WISDM	Wearable	50.0	66.3[b]	**70.3**±1.5	+4.0
SSC	EEG	58.0	63.5[c]	**66.2**±0.2	+2.7
MFD	Industrial	77.5	92.8[a]	**96.1**±0.1	+3.3
Mean		64.4	80.2	**82.7**	+2.6

[a]DIRT-T [b]CoTMix [c]MMDA

matches or exceeds the eICU-native LSTM on four of five tasks, surpassing both native references on AKI and LoS (Table 1). On AKI, adapted predictions reach 91.14 AUROC, above both the eICU-native (90.2) and MIMIC-IV-native (89.7) LSTMs, with larger AUCPR gains of $+16.1$ points ($56.8 \rightarrow 72.9$). On Sepsis (1.1% positive rate), the adapter reaches 77.76 AUROC, above the eICU-native LSTM (74.0); the margin over DA baselines widens to $3.7\times$ (Deep CORAL $+1.67$ vs. ours $+6.17$), since alignment-based methods spend gradient capacity on the domain objective, leaving little for the sparse task signal. LoS extends the approach to regression at 37.2 h MAE, beating both native references (39.2/40.6 h), and Mortality reaches 85.74 AUROC, exceeding the eICU-native LSTM (85.5).

Adapted inputs move *away* from the target distribution (Wasserstein $+3$–$5\times$) yet predictions improve, because the adapter disrupts forward-fill imputation artifacts (Appendix F). On HiRID → MIMIC-IV, gains grow: AKI and Sepsis each exceed the eICU best by over 0.02 AUROC (Sepsis reaches 79.8), and LoS drops to 35.4 h MAE (Appendix C.3). An LSTM-trained adapter also transfers zero-shot to frozen GRU and TCN predictors, retaining 47–86% of the LSTM improvement (Appendix D).

3.3. Generality: Non-Medical Time-Series Benchmarks

To test whether input-space adaptation generalizes beyond clinical data, we evaluate on the AdaTime benchmark (Ragab et al., 2023): a pretrained 1D-CNN is frozen; only the adapter is trained. Despite this strict constraint, under AdaTime's exact training recipe, the adapter outperforms all eleven published end-to-end DA methods on all five datasets (Table 2), mean $+2.6$ Macro-F1, with $+2.7$ F1 on sleep-stage EEG (SSC). Input-space adaptation is not limited to clinical data.

4. Conclusion

We introduced input-space domain adaptation: adapt the data, not the model. A single method outperforms every comparable baseline: four frozen-backbone DA methods (by 1.3–$3.7\times$), affine renormalization, end-to-end LSTM fine-tuning, three end-to-end clinical DA methods on the classification tasks (Appendix B), and all eleven published end-to-end methods on AdaTime under its own training recipe, while matching or exceeding the eICU-native LSTM on four of five clinical tasks and surpassing the MIMIC-IV-native LSTM on AKI and LoS. Because the predictor's weights never change, its SaMD clearance (U.S. Food and Drug Administration, 2021) remains intact.

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A. Dataset Statistics

Table 3. ICU databases used in this study. All data is preprocessed via the YAIB benchmark (van de Water et al., 2024) with standardized cohort definitions, feature extraction, and train/validation/test splits.

Dataset	Role	Stays	Hosp.	Country	Source
eICU	Source	200,859	208	US	Telehealth net.
MIMIC-IV	Target	73,181	1	US	Academic med. ctr.
HiRID	Source 2	33,905	1	Switz.	Univ. hospital

The 48 clinical features include vital signs (heart rate, blood pressure, SpO_2, temperature, respiratory rate) and laboratory values (metabolic panels, blood gases, liver enzymes, cardiac markers, coagulation studies). The 4 static features encode patient demographics (age, sex, height, weight). KF uses 192 additional generated features: cumulative minimum, maximum, mean, and count statistics for each of the 48 clinical features, which must maintain internal consistency after adaptation. LoS omits missing indicators because the YAIB LoS LSTM was trained on a 52-feature schema.

Table 4. Clinical prediction tasks. MI = missing indicator features (binary flags for whether a value was measured or forward-filled). Per-timestep tasks produce a prediction at every hourly timestep; per-stay tasks produce a single prediction per ICU stay.

Task	Type	Label	Granularity	Feature Dim	Pos. Rate
Mortality24	Classification	ICU death within 24 h	Per-stay	96 (48 + 48 MI)	∼5.5%
AKI	Classification	KDIGO Stage ≥1	Per-timestep	100 (48 + 48 MI + 4 static)	∼12%
Sepsis	Classification	Sepsis-3 criteria	Per-timestep	100 (48 + 48 MI + 4 static)	∼1.1%
LoS	Regression	Hours remaining	Per-timestep	52 (48 + 4 static)	N/A
KF	Regression	Serum creatinine (mg/dL)	Per-stay	292 (48 + 48 MI + 192 gen + 4 static)	N/A

Table 5. Hyperparameters for the RETRADAPTER architecture.

Hyperparameter	Value
d_{model}	128
d_{latent}	128
n_{heads}	8
d_{ff}	512
$n_{\text{enc_layers}}$	4
$n_{\text{dec_layers}}$	3
$n_{\text{cross_layers}}$	2 (Mort., Sepsis) / 3 (AKI, LoS, KF)
k (neighbors)	16
W (retrieval window)	6 timesteps (matches 6-hour clinical reassessment cycle)
Learning rate	5×10^{-4}
Optimizer	AdamW
Batch size	16
Phase 1 pretrain epochs	10–50 (task-dependent)
Phase 2 training epochs	30–100 (early stopping)
$\lambda_{\text{fidelity}}$	1.0
λ_{range}	0.1
Memory refresh (R)	5 epochs
Dropout	0.2

B. Implementation Details

Task-specific settings. Temporal attention mode is set to *causal* for per-timestep tasks (AKI, Sepsis, LoS) to prevent information leakage from future timesteps, and *bidirectional* for per-stay tasks (Mortality, KF) where the full sequence is available at prediction time. Cross-domain normalization (an affine transform matching per-feature mean and variance to target statistics, computed once from training data) is applied to all tasks. Phase 1 self-retrieval is optionally enabled: during autoencoder pretraining, a memory bank is built from the encoder's own target representations and provided to cross-attention blocks, avoiding degenerate zero-context initialization. We enable self-retrieval for all tasks in this work. Variable-length batching is used for per-timestep tasks and disabled for per-stay tasks.

Hardware. Primary development hardware: NVIDIA V100S-32GB. Multi-seed validation also used A6000-48GB, RTX 3090-24GB, L40S-48GB, and A100-40GB (SLURM cluster). Cross-server variance is within ±0.003 AUROC for all tasks.

Model sizes. The retrieval adapter contains 2.59M parameters (Mortality, Sepsis; $n_{\text{cross}} = 2$) to 2.88M parameters (KF; $n_{\text{cross}} = 3$). Frozen LSTM baselines range from 170K (Sepsis) to 1.42M (AKI) parameters. DA baselines: DANN 350K, Deep CORAL 251K, CoDATS 428K trainable parameters.

AdaTime configuration. The same RETRADAPTER architecture is used with reduced capacity: $d_{\text{model}} = 64$, $d_{\text{latent}} = 64$, $d_{\text{ff}} = 128$, $n_{\text{enc_layers}} = 2$, $n_{\text{dec_layers}} = 1$, $n_{\text{cross_layers}} = 2$–3. The frozen backbone is AdaTime's standard 3-block 1D-CNN (∼200K parameters). Adapter size: 100K–450K parameters depending on dataset. All AdaTime runs (ours and every baseline) follow the benchmark's exact recipe: 40-epoch budget, Adam with $\beta = (0.5, 0.99)$, weight decay 10^{-4}, last-epoch model, no source-validation split.

AdaTime end-to-end baselines. Ragab et al. (2023) benchmark eleven state-of-the-art end-to-end DA methods: DANN, Deep CORAL, CDAN, DSAN, HoMM, MMDA, CoDATS, AdvSKM, DIRT-T, CoTMix, and SASA. We also run CLUDA (Ozyurt et al., 2023) and RAINCOAT (He et al., 2023) under the same protocol. For each dataset, Table 2 reports the best of these published methods.

End-to-end DA on clinical tasks. As a stress test for the frozen-predictor assumption, we also ran three end-to-end DA methods that retrain the full LSTM on each clinical task: CLUDA (Ozyurt et al., 2023), RAINCOAT (He et al., 2023), and ACON (Liu et al., 2024). All three were swept over published hyperparameters and trained under right-padded YAIB data (matching the frozen-baseline protocol; left-padding inflated per-timestep AUROC and was excluded). Table 6 reports the best hyperparameter per method per task. RETRADAPTER, despite never touching the predictor, beats the best end-to-end DA on all three tasks.

C. Extended Results

C.1. AUCPR Results (eICU → MIMIC)

AKI AUCPR gains (+16.08) are nearly 3× the AUROC gains (+5.56): adaptation disproportionately improves pre-

Table 6. End-to-end DA baselines on clinical tasks (right-padded, best hyperparameter per method). AUROC ($\times 100$, ↑). Δ vs. the frozen source-only baseline.

Method	Mortality	AKI	Sepsis
Frozen source-only	80.79	85.58	71.59
E2E RAINCOAT	81.52	89.55	76.24
E2E CLUDA	81.81	87.35	76.61
E2E ACON	82.23	70.79	75.30
RETRADAPTER (ours, frozen)	**85.74**	**91.14**	**77.76**

Table 7. AUCPR results ($\times 100$) for classification tasks. Bootstrap 95% CIs (500 replicates).

	Frozen Baseline	RETRADAPTER [95% CI]	Δ
Mortality	29.67	34.88 [32.45, 37.36]	+5.21
AKI	56.78	72.86 [72.65, 73.06]	+16.08
Sepsis	2.98	5.32 [5.13, 5.51]	+2.34

cision at clinically relevant operating points.

C.2. Regression Extended Results (eICU → MIMIC)

Table 8. Regression results across multiple metrics. Bootstrap 95% CIs (500 replicates). MAE and RMSE: lower is better (↓); R^2: higher is better (↑).

Task	Metric	Baseline	RETRADAPTER [95% CI]	Δ
LoS	MAE (↓)	0.2527	0.2331 [.2329, .2333]	−0.0196
LoS	RMSE (↓)	0.3123	0.3004 [.3001, .3007]	−0.0119
LoS	R^2 (↑)	0.1130	0.1794 [.1782, .1806]	+0.0664
KF	MAE (↓)	0.0330	0.0256 [.0249, .0262]	−0.0074
KF	RMSE (↓)	0.0600	0.0477 [.0451, .0503]	−0.0123
KF	R^2 (↑)	0.7489	0.8411 [.8266, .8553]	+0.0922

Table 8 reports the primary retrieval variant for each task. The best task-specific variants (starred in Table 1) go further: LoS without MMD alignment achieves MAE 0.2212 (approximately 37.2 hours when denormalized), a 5.3-hour reduction from the frozen baseline's 42.5 hours and 2.0 hours below the eICU-native LSTM (39.2h); KF without fidelity and without MMD achieves normalized MAE 0.0239 ($\Delta = -0.0091$, 0.292 mg/dL denormalized). KF R^2 improves from 0.749 to 0.841, nearly doubling the explained variance in serum creatinine predictions. Dropping the fidelity loss removes fidelity gradient dominance in KF's 292-feature cumulative-statistics regime.

C.3. HiRID → MIMIC Extended Results

HiRID results show that the adapter works across source domains with very different data collection practices (Swiss vs. US ICUs). AKI and Sepsis improvements from HiRID (+0.078 AUROC each) exceed those from eICU (+0.056 and +0.062), possibly because greater domain distance provides more signal for adaptation. KF is the only non-significant result, likely due to HiRID's smaller cohort size (33,905 vs. 200,859 stays) providing insufficient coverage of the 292-feature KF schema.

Table 9. HiRID → MIMIC-IV results. Bootstrap 95% CIs (500 replicates). All $p < 0.001$ except KF ($p = 0.482$, n.s.).

Task	Metric	Δ vs Frozen [95% CI]	p
Mortality	AUROC	+0.0488 [+0.0284, +0.0711]	<0.001
Mortality	AUCPR	+0.0502 [+0.0167, +0.0859]	0.002
AKI	AUROC	+0.0776 [+0.0745, +0.0807]	<0.001
AKI	AUCPR	+0.1471 [+0.1414, +0.1531]	<0.001
Sepsis	AUROC	+0.0777 [+0.0707, +0.0841]	<0.001
Sepsis	AUCPR	+0.0525 [+0.0471, +0.0584]	<0.001
LoS	MAE	−0.0452 [−0.0497, −0.0410]	<0.001
KF	MAE	+0.0000 [−0.0009, +0.0010]	0.482

C.4. Seed Variance

Table 10. Multi-seed AUROC variance ($\times 100$). Spread = max − min across seeds.

Task	Config	Seeds	Mean Δ	Spread
AKI	V5, $n_{\text{cross}} = 3$	5	+5.47	0.27
Mortality	V4+MMD	3	+4.56	0.29
Sepsis	V4+MMD	3	+4.94	2.03

AKI and Mortality show low variance (spread <0.3 AUROC points), both well within noise. AKI's 5-seed validation spans 3 physically distinct GPU servers (V100S, A6000, RTX 3090), with cross-server variance adding only 0.2 points. Sepsis exhibits higher variance (spread 2.03) due to its 1.1% label density; this is inherent to the task rather than method-specific, as DA baselines show comparable variance.

D. Architecture-Agnostic Transfer

To test whether the adapter learns a domain-invariant input mapping rather than LSTM-specific artifacts, we apply an LSTM-trained adapter *zero-shot* to frozen GRU and TCN architectures (no retraining).

Table 11. Architecture-agnostic transfer: LSTM-trained adapter applied zero-shot to frozen GRU and TCN predictors. AUROC improvement ($\Delta \times 100$) over each architecture's frozen baseline. Percentage indicates fraction of LSTM gain retained.

Task	LSTM Δ	GRU Δ (%)	TCN Δ (%)
Mortality	+4.76	+4.04 (85%)	+3.42 (72%)
AKI	+5.56	+3.11 (56%)	+3.16 (57%)
Sepsis	+5.12	+2.40 (47%)	+4.42 (86%)

All six source–architecture combinations show positive transfer, with gains ranging from 47% to 86% of the LSTM

improvement. The strongest retention occurs for TCN on Sepsis (86%) and GRU on Mortality (85%); the input-space corrections appear largely architecture-independent. This supports the interpretation that the adapter resolves distributional mismatches (e.g., forward-fill imputation artifacts) at the input level, producing features that benefit any predictor trained on target-domain data.

E. Ablation Study

We ablate the main components of the retrieval adapter. Each row modifies exactly one setting from the control configuration.

Table 12. Component ablation on AKI (eICU → MIMIC-IV, AUROC ×100). Each row removes exactly one component from the full retrieval adapter ($n_{\text{cross}} = 3$).

Configuration	Δ AUROC	Effect
Full method (control)	+5.48	—
No retrieval (remove memory bank)	+4.88	−0.60
No feature gate	+5.02	−0.46
No MMD alignment	+5.19	−0.29
No fidelity loss	**+5.70**	+0.22
No target normalization	+5.43	−0.05

Table 13. Component ablation on Sepsis (eICU → MIMIC-IV, AUROC ×100). Control: full retrieval adapter with $n_{\text{cross}} = 2$.

Configuration	Δ AUROC	Effect
Full method (control)	+4.99	—
No smoothness loss	+4.75	−0.24
Oversampling (20×)	−3.31	−8.30
$n_{\text{cross}} = 3$ (vs 2)	+4.35	−0.64

Key findings. *Pretraining is essential*: removing Phase 1 (the autoencoder pretrain) collapses every classification task, with Mortality falling 10 AUROC points below the unadapted frozen baseline. *Retrieval provides consistent benefit*: removing the memory bank reduces AKI by 0.60 AUROC and all other tasks show similar degradation. *Oversampling is harmful for sparse labels*: 20× oversampling of positive examples causes catastrophic performance loss on Sepsis, likely due to gradient bias from repeated rare-event samples.

F. Interpretability: Imputation vs. ACF Change

G. Systematic Method Comparison

Table 14 compares representative methods from five categories across seven dimensions that collectively define the desiderata for clinical time-series domain adaptation with a frozen predictor. No prior method satisfies all seven. Representation-space (R) and weight-space (W) methods are architecturally coupled to the model's internals and cannot be formulated without access to intermediate representations or weight tensors, violating the frozen-model constraint. Input-space methods (VPT, Time-LLM, TATO) keep the model frozen but use input-independent perturbations (VPT, Time-LLM) or handcrafted transforms (TATO), limiting their capacity for heterogeneous clinical data where dozens of features shift simultaneously in different ways. kNN-MT (Khandelwal et al., 2021) is closest in spirit (frozen model, retrieval, per-timestep conditioning) but operates in output space (interpolating token distributions) rather than input space. RETRADAPTER is the only method combining a fully frozen predictor, learned input-space adaptation, sample-conditional neural transformation, target-domain retrieval, variable-length support, and evaluation across five clinical tasks.

H. Computational Cost

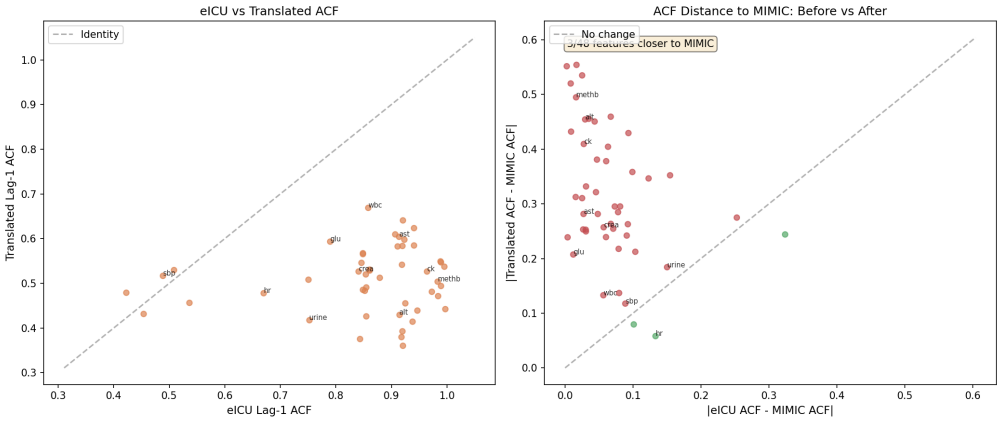
Table 15 reports single-seed training time per task. Phase 1 checkpoints are reusable, amortizing pretraining cost across experiments.

Parameter counts. Adapter: 2.59–2.88M parameters depending on task (dominated by 7 AxialBlocks at 265K each, plus 2–3 cross-attention blocks). Frozen LSTMs: 170K (Sepsis) to 1.42M (AKI). The adapter-to-model ratio ranges from 2.0× (AKI) to 15.2× (Sepsis), but the adapter operates in input space and never modifies the frozen model's parameters.

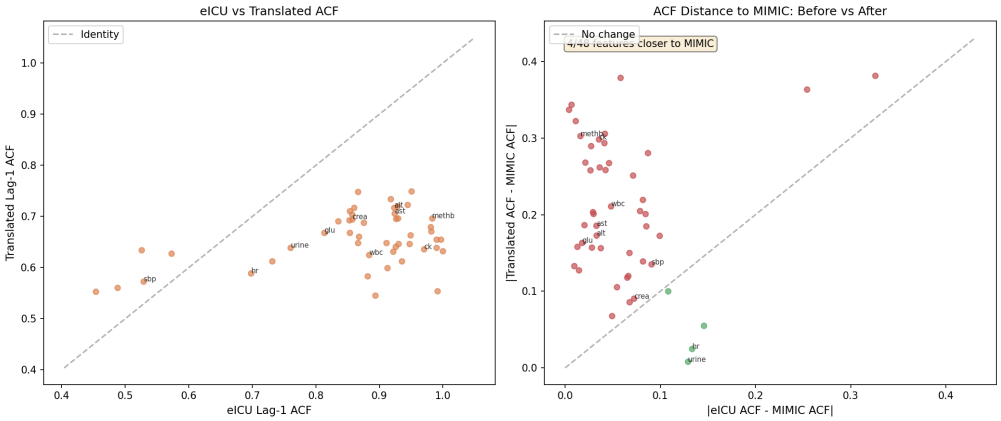
Inference overhead. At test time, the memory bank is pre-built once (∼6 minutes for AKI on V100S). Per-stay inference latency is ∼15.5 ms (AKI, including k-NN retrieval from a 424K-window bank), representing ∼9× overhead over the frozen LSTM alone; this is acceptable for clinical decision support where predictions are computed per hour.

Total compute budget. Full development required ∼2,400 GPU-hours across ∼700 experiments (architecture search, ablation studies, multi-seed validation, DA baselines, HiRID experiments). Minimum reproduction of the best configuration per task (single seed, all five EHR tasks plus five AdaTime datasets): ∼107 GPU-hours on V100S-32GB. Hardware: NVIDIA V100S-32GB (primary), A6000-48GB, RTX 3090-24GB, L40S-48GB, and A100-40GB (SLURM cluster).

AdaTime. Per-dataset training: ∼21 h across 10 DA scenarios × 5 seeds. Total AdaTime compute: ∼108 GPU-hours for full 5-seed validation across all 5 datasets.



(a) AKI (ρ=0.81)



(b) Sepsis (ρ=0.74)

Figure 2. Feature imputation fraction vs. adaptation-induced autocorrelation change (ΔACF). Laboratory values (high imputation, upper quadrant) undergo large temporal restructuring; continuously-monitored vitals (lower left) are preserved. Spearman ρ indicates the correlation between imputation fraction and ACF change, both $p < 10^{-9}$.

Table 14. Systematic comparison of domain adaptation and frozen-model methods across key desiderata. **Model Frozen**: entire target predictor weights fixed (✓), partially fixed (P), or trainable (✗). **Adapt. Space**: Input (I), Representation (R), Weight (W), or Output (O) space. **Sample-Cond.**: transformation depends on the input instance. **Retrieval**: uses a domain-specific datastore for per-instance conditioning. **Var.-Len. TS**: handles variable-length time series natively. **Clinical Tasks**: number of distinct clinical prediction tasks evaluated (0 = non-clinical only). Only our method satisfies all desiderata simultaneously.

Method	**Model Frozen**	**Adapt. Space**	**Sample-Cond.**	**Retrieval**	**Var.-Len. TS**	**Clinical Tasks**
Clinical DA						
VRADA (Purushotham et al., 2017)	✗	R	✓	✗	✓	1
OTTEHR (OTTEHR Authors, 2025)	✗	R	✗	✗	✗	3
Mutnuri et al. (Mutnuri et al., 2024)	✗	W	✓	✗	✓	3
General TS-DA						
CoDATS (Wilson et al., 2020)	✗	R	✓	✗	✗	0
RAINCOAT (He et al., 2023)	✗	R	✓	✗	✗	0
CLUDA (Ozyurt et al., 2023)	✗	R	✓	✗	✗	0
Frozen-model / Input-space						
VPT (Jia et al., 2022)	✓	I	✗	✗	✗	0
Time-LLM (Jin et al., 2024)	✓	I	✗	✗	✗	0
TATO (Qiu et al., 2026)	✓	I	P	✗	✓	0
Retrieval-augmented						
kNN-MT (Khandelwal et al., 2021)	✓	O	✓	✓	✓	0
TS-RAG (Ning et al., 2025)	✗	R	✓	✓	✓	0
Foundation model						
ICareFM (Burger et al., 2025)	✗	W	✓	✗	✓	4
RETRADAPTER (Ours)	✓	**I**	✓	✓	✓	**5**

Table 15. Training time per task (single seed, V100S-32GB). Phase 1 checkpoints are reusable across experiments with matching architecture.

Task	**Phase 1 (h)**	**Phase 2 (h)**	**Total (h)**
Mortality	4.5	9.0	13.5
AKI	6.8	15.8	22.5
Sepsis	6.8	13.5	20.3
LoS	7.5	15.0	22.5
KF	2.5	5.0	7.5